ECN 594: Midterm Review Notes

Nicholas Vreugdenhil

1 Master Formula Sheet

Topic	Formula	Notes
Demand Estimation		
Berry inversion	$\delta_{jt} = \ln(s_{jt}) - \ln(s_{0t})$	Go from shares to mean utility
Own-price elasticity	$\eta_{jj} = \alpha p_j(1 - s_j)$	$\alpha < 0 \Rightarrow \eta_{jj} < 0$
Cross-price elasticity	$\eta_{jk} = -\alpha p_k s_k$	Same for all j (IIA problem)
Consumer surplus	$CS = \frac{1}{\lvert\alpha\rvert} \ln \left[\sum_j \exp(\delta_j)\right]$	Include outside option ($\delta_0 = 0$)
Monopoly Pricing		
Lerner index	$L = \frac{p-MC}{p} = \frac{1}{\lvert\varepsilon\rvert}$	Markup = inverse elasticity
Optimal price	$p = \frac{MC}{1-1/\lvert\varepsilon\rvert}$	Rearranged Lerner
Back out MC	$MC = p\left(1 - \frac{1}{\lvert\varepsilon\rvert}\right)$	Given p and ε
Two-Part Tariffs (Homogeneous)		
Optimal tariff	$p^* = MC, \quad F^* = CS(p^*)$	Price at cost, extract CS via fee
Self-Selection		
IR binds for L	$p_S = v_L^S$	L gets zero surplus
IC binds for H	$p_F = v_H^F - v_H^S + p_S$	H indifferent between F and S

2 Demand Estimation

2.1 Key Concepts

- **Utility:** $u_{ijt} = \delta_{jt} + \varepsilon_{ijt}$ where $\delta_{jt} = x_{jt}\beta + \alpha p_{jt} + \xi_{jt} \quad (\alpha < 0)$
- **Shares:** $s_{jt} = \frac{\exp(\delta_{jt})}{1+\sum_k \exp(\delta_{kt})}, \quad s_{0t} = \frac{1}{1+\sum_k \exp(\delta_{kt})}$
- **Price endogeneity:** High $\xi_{jt} \rightarrow$ high price $\Rightarrow$ OLS biases α toward zero (less negative)
- **IVs:** Cost shifters, Hausman (prices elsewhere), BLP (rival characteristics)
- **IIA problem:** Cross-elasticity η_{jk} doesn't depend on j $\Rightarrow$ unrealistic substitution
- **Demographics help:** $u_{ijt} = \delta_{jt} + \sum_{k,\ell} \pi_{k\ell} D_{i\ell} x_{jt,k} + \varepsilon_{ijt} \Rightarrow$ richer substitution

2.2 Cookbook: Berry Inversion

1. Compute $\delta_j = \ln(s_j) - \ln(s_0)$ for each product
2. Interpretation: higher δ = more preferred (relative to outside option)

2.3 Cookbook: Elasticity Calculation

1. Own-price: $\eta_{jj} = \alpha p_j(1 - s_j)$. Since $\alpha < 0$, this is negative.
2. Cross-price: $\eta_{jk} = -\alpha p_k s_k$. Since $\alpha < 0$, this is positive (substitutes).
3. With small shares: $\eta_{jj} \approx \alpha p_j$ (higher price $\Rightarrow$ more elastic)

2.4 Cookbook: Consumer Surplus Change

1. Before: $CS^{\text{before}} = \frac{1}{|\alpha|} \ln(e^{\delta_0} + e^{\delta_1} + \ldots)$ where $\delta_0 = 0$
2. After: $CS^{\text{after}} = \frac{1}{|\alpha|} \ln(\text{sum over remaining products})$
3. Loss: $\Delta CS = CS^{\text{before}} - CS^{\text{after}}$

Common mistake: Using $\sum \exp(\delta_j)$ instead of $\ln[\sum \exp(\delta_j)]$

2.5 Demographic Interactions Model

- **Model:** $u_{ijt} = \delta_{jt} + \mu_{ijt} + \varepsilon_{ijt}$
- Mean utility: $\delta_{jt} = x_{jt}\beta + \alpha p_{jt} + \xi_{jt}$ (common to all consumers)
- Individual deviation: $\mu_{ijt} = \sum_{k,\ell} \pi_{k\ell} D_{i\ell} x_{jt,k}$ (e.g., income $\times$ price)
- **Advantage:** Richer substitution patterns—products that attract similar consumers are closer substitutes
- **Example:** If $\pi_{\text{inc,price}} > 0$, high-income consumers are less price-sensitive

2.6 Red Bus / Blue Bus Problem

- **Setup:** Car and Red Bus, equal shares. Introduce identical Blue Bus.
- **Logit predicts:** CS increases (adds an extra logit draw)
- **Reality:** No real welfare gain (it's the same bus!)
- **Fix:** Use observed post-introduction shares to back out δ's. The lower shares for each bus offset the extra logit gain.

2.7 Applications

- **Petrin (2002):** Valued minivan introduction at \$2.8B. Key insight: logit overstates CS gains; micro-data helps.
- **Nevo (2001):** Cereal demand. Demographics $\rightarrow$ richer substitution. Used to study market power.

3 Pricing and Price Discrimination

3.1 Monopoly Pricing

- **Lerner index:** $L = \frac{p - MC}{p} = \frac{1}{|\varepsilon|}$ (markup = inverse elasticity)
- **MC pricing:** $p = MC \Rightarrow$ DWL = 0, but firm needs subsidy if $F > 0$
- **AC pricing:** $p = AC \Rightarrow$ zero profit, but DWL > 0 (some consumers with WTP $\in (MC, AC)$ don't buy)

3.2 Cookbook: Selection by Indicators

1. Apply Lerner in each market: $\frac{p_i - MC}{p_i} = \frac{1}{|\varepsilon_i|}$
2. Solve: $p_i = \frac{MC}{1-1/|\varepsilon_i|}$
3. Result: higher price in more inelastic market

3.3 Cookbook: Two-Part Tariffs (Homogeneous Consumers)

1. Set $p^* = MC$ (maximizes total surplus)
2. Compute quantity: $q^* = D(p^*)$
3. Compute $CS = \frac{1}{2} \times q^* \times (\text{choke price} - p^*)$ for linear demand
4. Set $F^* = CS$ (extract all surplus via fee)
5. Profit $= F^*$ (margin on sales is zero)

3.4 Cookbook: Two-Part Tariffs (Heterogeneous Consumers)

1. At $p = 0$, compute CS_H and CS_L for each type
2. Strategy A: Set $F = CS_H \Rightarrow$ only H joins, profit $= F \times N_H$
3. Strategy B: Set $F = CS_L \Rightarrow$ both join, profit $= F \times (N_H + N_L)$
4. Compare profits to choose optimal F

3.5 Cookbook: Self-Selection Menu Design

Setup: Two types (H, L), two versions (Full, Stripped), WTP table given.

1. Set IR for L to bind: $p_S = v_L^S$ (L gets zero surplus)
2. Set IC for H to bind: $v_H^F - p_F = v_H^S - p_S \Rightarrow p_F = v_H^F - v_H^S + p_S$
3. Verify IC for L: $v_L^S - p_S \geq v_L^F - p_F$ (should hold by construction)
4. Profit $= (p_F - MC) \times N_H + (p_S - MC) \times N_L$

Key insight: H gets "information rent" $= v_H^S - v_L^S$; L gets zero surplus.

3.6 Cookbook: Checking a Given Menu

1. For each type, compute $CS = WTP - price$ for each option
2. Each type buys the option with highest CS (if ≥ 0)
3. If H buys Stripped instead of Full, IC is violated $\Rightarrow$ menu is suboptimal